

## Final Project Part 1 Group 3

### 1. Language name and brief description.

A programming language call ANVI (stands for America, Nepal, Vietnam, India and also stands for ANtiVirus, aiming to be a highly secure programming language). It is a beginner-friendly programming language designed for learning the fundamentals of programming language implementation and its syntax is inspired by C++, Java, and Python.

### 2. List of keywords.

| Keyword | Purpose                                    |
|---------|--------------------------------------------|
| let     | Declares a variable                        |
| if      | Starts a conditional statement             |
| else    | Provides an alternative conditional branch |
| wloop   | Starts a while loop                        |
| floop   | For loop                                   |
| func    | Defines a function                         |
| return  | Returns a value from a function            |
| display | Prints a value                             |
| T       | Boolean true value                         |
| F       | Boolean false value                        |

### 3. Operators and delimiters.

#### Arithmetic Operators

| Operator | Meaning  |
|----------|----------|
| +        | Addition |

|   |                  |
|---|------------------|
| - | Subtraction      |
| * | Multiplication   |
| / | Division         |
| % | Modulo/remainder |

### Comparison Operators

| Operator | Meaning                  |
|----------|--------------------------|
| ==       | Equal to                 |
| !=       | Not equal to             |
| <        | Less than                |
| >        | Greater than             |
| <=       | Less than or equal to    |
| >=       | Greater than or equal to |

### Assignment Operator

| Operator | Meaning                       |
|----------|-------------------------------|
| =        | Assigns a value to a variable |

### Delimiters

| Symbol | Purpose                                    |
|--------|--------------------------------------------|
| ( )    | Function calls, parameters, and conditions |
| { }    | Code blocks                                |
| [ ]    | Array indexing                             |
| ,      | Separates parameters or arguments          |
| ;      | Ends a statement                           |

#### 4. Variable declaration/assignment syntax.

Let count = 0;

This would assign a variable for Count to equal 0 to end this statement would use a “;”.

#### 5. Control-structure syntax.

Control structure tells the computer what to do. So from our prior example we will combine that with instructions for what the computer will do.

Let count = 0

```
If (count == 0)
```

```
{print ("count =" + count );
```

```
}
```

```
Else {print "end"}
```

This tells the computer what to do if count equals 0. Then if it is not equal to 0 then it tells the computer what else to do.

#### 6. Function syntax.

This stores information you can use.

```
Let sayHello() {print ("Hello") }
```

You can then use sayHello repeatedly throughout the code.

7. At least **three complete sample programs**.
8. Initial BNF/EBNF grammar.
9. Proposed additional/extension feature.
10. Initial responsibilities of each group member.